

APPENDIX 4 – FULL LIST OF 43 EQUATIONS

DATA SOVEREIGNTY AND DIGITAL RIGHTS

12 Organizations

EQUATION 1/43: DAIR (DISTRIBUTED AI RESEARCH INSTITUTE)

Location: USA | **Sector:** Community centered AI research institute

Geographic Scope: Global (distributed and embedded in communities)

Maturity: Est. 2021; emerging

Conditions and their mechanisms:

- **Decentralized Decision-Making** (Governance) *structures authority* by distributing research governance to affected communities rather than concentrating it in traditional institutional hierarchies
- **Community Capacity Building** (Resources) *builds agency* by investing in community-level AI skills and knowledge that enable meaningful participation
- **Participatory Technology Design** (Technology Practices) *facilitates deliberation* by centering affected communities in design processes

Condition interactions: Decentralized Decision-Making creates the governance architecture within which Community Capacity Building and Participatory Technology Design operate—without distributed authority, capacity building would lack channels for influence, and participatory design would lack decision-making power. Community Capacity Building provides the substrate for meaningful participation in both governance and design. Participatory Technology Design translates governance authority and community capacity into concrete technical choices.

Outcomes produced:

- **Researcher autonomy with distributed decision-making** (Democratic Practices): Decentralized Decision-Making *structures authority* that Community Capacity Building *enables* researchers to exercise
- **Small-scale community-driven AI models** (Democratic Practices): Participatory Technology Design *facilitates deliberation* translating community priorities into technical specifications
- **Enhanced community capacity for contextually appropriate AI** (Impact): Community Capacity Building *builds agency* creating lasting infrastructure for community-led AI development

Synthesis: DAIR demonstrates how decentralized governance, community investment, and participatory design combine to produce both internal democratic practices and external capacity-building impact. The case shows that challenging AI's concentrated power requires simultaneous transformation of governance structures (who decides), resource distribution (who has capacity), and design processes (how technology is shaped)—with each dimension reinforcing the others. Through the D-A-S framework, DAIR's design choices create

affordances for distributed AI development that structure new forms of sovereignty capacity for marginalized communities.

COMPACT EQUATION: For DAIR (Distributed AI Research Institute), DD (Governance) + CCB (Resources) + PTD (Technology P) = RAWD (Dem) + SCAM (Dem) + ECCT (Imp)

EQUATION 2/43: DETROIT COMMUNITY TECHNOLOGY PROJECT (CTC)

Location: USA | **Sector:** Advocacy group for digital rights

Geographic Scope: Built locally (in Detroit, Tennessee, LA, and NY) but now with a national span; principles the organ

Maturity: Est. 2011; reasonably mature

Conditions and their mechanisms:

- **Decentralized Decision-Making** (Governance) *structures authority* by distributing technology governance to neighborhood residents rather than institutional actors
- **Capacity Building and Technological Empowerment** (Resources) *enables self-governance* by developing local digital skills that allow communities to maintain and govern their own infrastructure
- **Community-Owned Infrastructure** (Technology Tools) *materializes control* by creating physical networks that communities actually own and operate

Condition interactions: Community-Owned Infrastructure provides the material foundation upon which Decentralized Decision-Making can operate—governance of infrastructure requires infrastructure to govern. Capacity Building and Technological Empowerment ensures communities can exercise the authority that Decentralized Decision-Making distributes. The three conditions create a virtuous cycle: infrastructure ownership motivates capacity building, capacity building enables effective governance, and effective governance strengthens infrastructure ownership.

Outcomes produced:

- **Community-driven collaborative decision-making** (Democratic Practices): Decentralized Decision-Making *structures authority* through collective processes
- **Local technical capacity enabling self-governance** (Democratic Practices): Capacity Building *enables self-governance* through skills development
- **Community-controlled digital infrastructure** (Democratic Practices): Community-Owned Infrastructure *materializes control* in physical networks
- **Portable community networking** (Community Value): Community-Owned Infrastructure *produces assets* that communities can deploy independently

Synthesis: Detroit CTC demonstrates how infrastructure ownership, skills development, and distributed governance combine to produce comprehensive community control over digital communications. The case shows that digital self-determination requires material foundations (owned infrastructure), human capacity (technical skills), and governance structures (distributed authority)—each necessary but insufficient alone. Detroit's approach creates

lasting sovereignty capacity by building all three dimensions simultaneously, ensuring communities can not only access but actually control their digital infrastructure.

COMPACT EQUATION: For Detroit Community Technology Project (CT, DD (Governance) + CBATE (Resources) + CI (Technology) = CCDM (Dem) + LTCE (Dem) + CDCA (Dem) + PCNE (Com)

Consolidation Notes:

- Merged 1 condition group(s)

EQUATION 3/43: DIGITAL EMPOWERMENT FOUNDATION

Location: India | **Sector:** Digital literacy

Geographic Scope: National

Maturity: Est. 2002; mature

Conditions and their mechanisms:

- **Community Capacity Building** (Org Info) *positions* the organization as a capacity-building intermediary bridging digital divides
- **Bidirectional Knowledge Exchange** (Partners) *structures exchange* by creating flows that strengthen both local communities and global networks
- **Community-Owned Infrastructure Development** (Technology Practices) *enables practice* by building village-level internet and language applications
- **Democratizing Access and Skills** (Resources) *democratizes skills* by removing barriers to digital and financial literacy
- **Inclusive Participation and Representation** (Social Context) *enables inclusion* by structuring participation to reach marginalized groups
- **Culturally-Embedded Technology Development** (Technology Tools) *embeds culture* by creating local language applications

Condition interactions: Bidirectional Knowledge Exchange connects local capacity building to global learning, ensuring that Democratizing Access and Skills draws on broader resources while Inclusive Participation and Representation ensures benefits reach marginalized groups. Community-Owned Infrastructure Development and Culturally-Embedded Technology Development translate these flows into tangible local assets. The conditions work as an integrated system: global connections feed local capacity, local capacity enables infrastructure development, and cultural embedding ensures relevance.

Outcomes produced:

- **Community technological capacity and expanded participation** (Community Value): Democratizing Access and Skills *democratizes access* creating widespread digital capacity
- **Village-owned infrastructure with local language applications** (Community Value): Community-Owned Infrastructure Development *enables practice* through locally-appropriate technology

- **Reciprocal knowledge exchange** (Balance): Bidirectional Knowledge Exchange *structures exchange* ensuring mutual benefit
- **Active leadership roles for marginalized groups** (Democratic Practices): Inclusive Participation and Representation *enables inclusion* in technology governance

Synthesis: Digital Empowerment Foundation demonstrates how global-local knowledge flows, skills democratization, and culturally-embedded infrastructure combine to produce community value and democratic practices in underserved contexts. The case shows that addressing digital divides requires not just connectivity but comprehensive capacity building that includes infrastructure, skills, cultural relevance, and governance inclusion. DEF's approach creates sovereignty capacity by ensuring communities have not just access to technology but the skills, culturally-appropriate tools, and governance voice to shape technology to their own purposes.

COMPACT EQUATION: For Digital Empowerment Foundation, CCB (Org Info) + BKE (Partners) + CID (Technology P) + DAAS (Resources) + IPAR (Social Conte) + CTD (Technology) = CTCB (Com) + VIIW (Com) + WDAF (Com) + RKES (Bal) + ALRF (Dem)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 4/43: FIRST NATIONS INFORMATION GOVERNANCE CENTRE (FNIGC)

Location: Canada | **Sector:** Advocacy network - indigenous data rights

Geographic Scope: National, but participate in global networks; Global South

Maturity: Est. 2010, but sovereignty work began in 1996; mature

Conditions and their mechanisms:

- **Multi-Network Presence and Translocal Engagement** (Governance) *connects networks* by positioning FNIGC across national and global Indigenous sovereignty movements
- **Strategic Institutional Engagement** (Politics) *legitimizes practice* by engaging colonial institutions on First Nations terms
- **Policy Influence and Systemic Change** (Politics) *directly produces* policy frameworks recognizing Indigenous data sovereignty
- **Indigenous Knowledge Systems Integration** (Purpose) *grounds governance* by anchoring data governance in Indigenous epistemologies
- **Strategic Navigation of Regulatory Landscapes** (Regulation) *navigates regulation* in hostile frameworks designed to privilege state control
- **Community Capacity Building** (Resources) *builds capacity* for First Nations to govern their own data
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating mechanisms for First Nations authority over their data
- **Indigenous Data Sovereignty** (Technology Tools) *instantiates sovereignty* through purpose-built data infrastructure

Condition interactions: Indigenous Knowledge Systems Integration provides the foundational orientation that shapes how all other conditions operate—without epistemological grounding, engagement with colonial institutions risks assimilation. Policy Influence and Strategic Institutional Engagement translate this grounding into external recognition, while Strategic Navigation of Regulatory Landscapes enables operation within hostile frameworks. Democratic Data Governance and Indigenous Data Sovereignty create the technical infrastructure, while Community Capacity Building ensures First Nations can actually exercise governance authority.

Outcomes produced:

- **Multi-scalar governance maintaining local control** (Impact): Multi-Network Presence *connects networks* while preserving community accountability
- **Policy frameworks recognizing Indigenous data sovereignty** (Impact): Policy Influence *directly produces* institutional change
- **Regional capacity for First Nations data sovereignty** (Impact): Community Capacity Building *builds capacity* at regional scale
- **First Nations-controlled data governance** (Democratic Practices): Democratic Data Governance *establishes control* through community-driven mechanisms
- **Strategic partnerships maintaining independence** (Democratic Practices): Strategic Institutional Engagement *legitimizes practice* on First Nations terms

Synthesis: FNIGC demonstrates how Indigenous epistemological grounding, strategic institutional engagement, and community capacity building combine to produce both policy impact and democratic data governance practices. The case shows that Indigenous data sovereignty requires simultaneous work across multiple fronts: establishing epistemological foundations, building technical infrastructure, developing community capacity, and transforming external policy frameworks—all while navigating hostile regulatory environments. FNIGC's success stems from holding these dimensions together, ensuring that policy victories are grounded in actual First Nations governance capacity and epistemological integrity.

COMPACT EQUATION: For First Nations Information Governance Cen, MPATE (Governance) + SIE (Politics) + PIASC (Politics) + IKSI (Purpose) + SNORL (Regulation) + CCB (Resources) + DDG (Technology P) + IDS (Technology) = MGML (Imp) + PFRI (Imp) + RCFF (Imp) + SPAF (Dem) + FNDG (Dem) + INEP (Dem) + FNSO (Dem)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 5/43: GLOBAL INDIGENOUS DATA ALLIANCE (GIDA)

Location: Spain | **Sector:** Advocacy network - indigenous data rights

Geographic Scope: Global network; Global South

Maturity: Est. 2013; mature

Conditions and their mechanisms:

- **Multi-Network Presence** (Partners) *enables coordination* by connecting isolated Indigenous data sovereignty movements globally
- **Policy Influence and Systemic Change** (Politics) *legitimizes frameworks* by securing international institutional adoption of Indigenous governance principles
- **Indigenous Knowledge Systems Integration** (Purpose) *grounds governance* by centering Indigenous worldviews in data governance design
- **Self-Determined Indigenous Data Sovereignty** (Regulation) *asserts sovereignty* independent of state frameworks
- **Embedding in Indigenous Sovereignty Movements** (Social Context) *embeds movement* by situating data governance within broader Indigenous self-determination struggles
- **Indigenous Data Sovereignty** (Technology Tools) *instantiates sovereignty* through tools designed for Indigenous governance

Condition interactions: Multi-Network Presence creates the infrastructure for global coordination, while Embedding in Indigenous Sovereignty Movements ensures data governance remains connected to broader self-determination struggles. Indigenous Knowledge Systems Integration provides the shared epistemological foundation that makes transnational coordination possible without bureaucratic hierarchy. Policy Influence and Self-Determined Indigenous Data Sovereignty represent complementary strategies—engaging international institutions while maintaining independence from state frameworks.

Outcomes produced:

- **Transnational Indigenous coordination** (Balance): Multi-Network Presence *enables coordination* amplifying local sovereignty efforts
- **Indigenous governance frameworks adopted internationally** (Impact): Policy Influence *legitimizes frameworks* in international institutions
- **Global framework for Indigenous data governance** (Democratic Practices): Indigenous Knowledge Systems Integration *grounds governance* in collective Indigenous values
- **Indigenous collective authority independent of state control** (Democratic Practices): Self-Determined Indigenous Data Sovereignty *asserts sovereignty* beyond state frameworks

Synthesis: GIDA demonstrates how global Indigenous network coordination, epistemological grounding, and strategic policy engagement combine to produce transnational impact while maintaining Indigenous self-determination. The case shows that Indigenous data sovereignty at the global level requires building networks that amplify local efforts without creating bureaucratic hierarchies, engaging international institutions while maintaining independence from state frameworks, and grounding transnational coordination in shared Indigenous epistemologies rather than Western governance models. GIDA's CARE Principles represent this synthesis—principles derived from Indigenous worldviews that have achieved international institutional recognition.

COMPACT EQUATION: For Global Indigenous Data Alliance (GIDA), MP (Partners) + PIASC (Politics) + IKSI (Purpose) + SRTDS (Regulation) + EISM (Social Conte) + IDS (Technology) = TICA (Bal) + TISN (Bal) + IGFA (Imp) + GFFI (Dem) + ICAO (Dem)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 6/43: MAIAM NAYRI WINGARA

Location: Australia | **Sector:** Data rights - cultural preservation

Geographic Scope: National, but participate in global networks; Global South

Maturity: Est. 2018; semi-mature

Conditions and their mechanisms:

- **Value Reconceptualization** (Purpose) *reframes data* by challenging colonial frameworks that treat Indigenous data as public property and reconceptualizing data as collective Indigenous asset
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating governance mechanisms that center Indigenous agency rather than framing Indigenous Australians as problems

Condition interactions: Value Reconceptualization provides the foundational reframing that enables Democratic Data Governance to operate on different terms—without challenging the assumption that Indigenous data is public property, governance mechanisms would reproduce colonial relationships. Together, they create conditions for data governance that centers Indigenous agency and collective benefit.

Outcomes produced:

- **Community-centered valuation of data as collective Indigenous asset** (Community Value): Value Reconceptualization *redefines value* in terms of collective Indigenous benefit
- **Aboriginal and Torres Strait Islander data governance centering Indigenous agency** (Democratic Practices): Democratic Data Governance *establishes control* through Indigenous-led mechanisms

Synthesis: MAIAM NAYRI WINGARA demonstrates how value reconceptualization and democratic data governance combine to challenge colonial data practices and center Indigenous agency. The case shows that transforming Indigenous data governance requires not just new governance mechanisms but fundamental reconceptualization of what data is and whom it serves—shifting from data as public resource to be exploited to data as collective Indigenous asset to be governed by and for Indigenous peoples. This foundational reframing enables governance practices that genuinely center Indigenous agency rather than reproducing colonial relationships under new technical forms.

COMPACT EQUATION: For MAIAM NAYRI WINGARA, VR (Purpose) + DDG (Technology P) = CVOD (Com) + AATS (Dem)

EQUATION 7/43: MASAKHANE

Location: Africa | **Sector:** Technology research for digital sovereignty

Geographic Scope: Regional; Global South
Maturity: Est. 2019; emerging

Conditions and their mechanisms:

- **Inclusive Expertise Integration** (Partners) *integrates knowledge* by bridging technical AI expertise with linguistic community participation
- **Bidirectional Knowledge Exchange** (Partners) *exchanges knowledge* by creating flows between academic researchers and local linguistic communities
- **Value-Based Resource Allocation** (Resources) *aligns resources* with community linguistic priorities rather than external metrics
- **Community Capacity Building** (Resources) *builds capacity* for African AI research beyond Western institutions
- **Decolonial Technology Development** (Problem) *counters domination* by motivating work that challenges Western AI dominance
- **Cultural Sovereignty and Knowledge Preservation** (Purpose) *directs preservation* toward African linguistic self-determination
- **Culturally-Embedded Technology Development** (Technology Practices) *embeds culture* by grounding NLP development in language-specific community needs

Condition interactions: Decolonial Technology Development provides the orienting challenge that motivates all other conditions. Inclusive Expertise Integration and Bidirectional Knowledge Exchange structure relationships between technical experts and communities in non-extractive ways. Community Capacity Building ensures benefits remain in African contexts. Culturally-Embedded Technology Development translates these relationships into technology that serves language-specific needs.

Outcomes produced:

- **Productive integration of expertise with community knowledge** (Democratic Practices): Inclusive Expertise Integration *integrates knowledge* across technical and community domains
- **Open-access datasets for African languages** (Community Value): Bidirectional Knowledge Exchange *circulates knowledge* as commons
- **Culturally appropriate technology with community ownership** (Community Value): Culturally-Embedded Technology Development *embeds culture* in technical outputs
- **University partnerships ensuring local benefits** (Balance): Bidirectional Knowledge Exchange *structures exchange* for mutual benefit
- **African language technological capacity** (Impact): Community Capacity Building *builds capacity* for ongoing African-led development

Synthesis: Masakhane demonstrates how decolonial purpose, inclusive expertise integration, and community capacity building combine to produce both valuable technology commons and lasting African AI research capacity. The case shows that countering Western AI dominance requires not just creating African language technology but fundamentally restructuring how AI research is conducted—building non-extractive relationships between technical experts and linguistic communities, ensuring community ownership of outputs, and

developing capacity for ongoing African-led research. Masakhane's distributed, community-driven model creates sovereignty capacity that persists beyond any single project.

COMPACT EQUATION: For Masakhane, IEI (Partners) + BKE (Partners) + VRA (Resources) + CCB (Resources) + DTD (Problem) + CSAKP (Purpose) + CTD (Technology P) = PIOT (Dem) + ODAT (Com) + CATW (Com) + UPEL (Bal) + ALTC (Imp) + ALNC (Imp)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 8/43: MONLAM AI

Location: Tibetan Diaspora | **Sector:** Software / Technology services

Geographic Scope: Digital - Global - for diaspora

Maturity: Est. 2023; Emerging

Conditions and their mechanisms:

- **Community-Controlled Technology Development** (Partners) *protects sovereignty* by ensuring external technical partnerships remain under monastic and scholarly oversight
- **Cultural Heritage Preservation through Technology** (Problem) *counters threat* by responding to political suppression and cultural erasure with technological preservation
- **Cultural Sovereignty and Knowledge Preservation** (Purpose) *advances sovereignty* through explicit orientation toward Tibetan linguistic and cultural self-determination
- **Culturally-Embedded Technology Development** (Technology Tools) *embeds culture* by integrating technology development with Buddhist scholarly traditions

Condition interactions: Cultural Heritage Preservation through Technology provides urgency—the threat of cultural erasure motivates action. Cultural Sovereignty and Knowledge Preservation orients that action toward sovereignty rather than mere preservation. Community-Controlled Technology Development ensures external partnerships serve rather than compromise community control. Culturally-Embedded Technology Development translates these orientations into technology that embodies Buddhist scholarly traditions.

Outcomes produced:

- **Technology development aligned with Tibetan cultural values** (Balance): Community-Controlled Technology Development *protects sovereignty* in external partnerships
- **Tibetan linguistic sovereignty through community-controlled AI** (Impact): Cultural Sovereignty and Knowledge Preservation *advances sovereignty* at technological level
- **Tibetan language technology preserving cultural heritage** (Impact): Culturally-Embedded Technology Development *preserves culture* through technical artifacts

- **Tibetan NLP tools integrated with monastic institutions** (Community Value): Culturally-Embedded Technology Development *embeds culture* in institutional contexts

Synthesis: Monlam AI demonstrates how cultural threat response, sovereignty-oriented purpose, and community-controlled partnerships combine to produce technology that serves Tibetan cultural preservation under conditions of political suppression. The case shows that cultural and linguistic sovereignty in threatened contexts requires technology that does more than digitize heritage—it must be controlled by community institutions, developed according to cultural protocols, and oriented toward sovereignty rather than mere preservation. Monlam's integration with monastic institutions ensures that technology serves cultural continuity as defined by Tibetan traditions themselves.

COMPACT EQUATION: For Monlam AI, CTD (Partners) + CHPTT (Problem) + CSAKP (Purpose) + CTD (Technology) = TDAW (Bal) + TLST (Imp) + TLTP (Imp) + TNTI (Com)

EQUATION 9/43: TE HIKU MEDIA

Location: New Zealand | **Sector:** Media organization nurturing Maori language and culture
Geographic Scope: National
Maturity: Est 1990; mature

Conditions and their mechanisms:

- **Indigenous Data Sovereignty** (Partners) *protects sovereignty* by structuring partnerships that maintain Māori control over language data despite corporate interest
- **Alternative Value Definition** (Problem) *reconceptualizes value* by challenging narrow economic definitions that exclude cultural and social dimensions
- **Indigenous Knowledge Systems Integration** (Purpose) *grounds governance* by anchoring AI development in Māori values and tikanga
- **Cultural Sovereignty and Knowledge Preservation** (Purpose) *directs preservation* toward linguistic revitalization as self-determination
- **Legal Innovation for Community Control** (Regulation) *formalizes protection* by creating licensing frameworks that prevent appropriation of Indigenous language data
- **Culturally-Embedded Technology Development** (Technology Tools) *embeds culture* by building AI that reflects Māori protocols and serves cultural purposes

Condition interactions: Alternative Value Definition reframes what counts as success, enabling rejection of extractive partnerships. Indigenous Knowledge Systems Integration and Cultural Sovereignty provide the orientation that shapes technology development. Legal Innovation for Community Control creates protective frameworks. Indigenous Data Sovereignty structures external relationships. Culturally-Embedded Technology Development translates these into world-leading technology that serves Māori purposes.

Outcomes produced:

- **Independent technology development retaining community control** (Balance): Indigenous Data Sovereignty *protects sovereignty* against corporate appropriation

- **Cultural sovereignty and linguistic preservation** (Community Value): Cultural Sovereignty and Knowledge Preservation *directs preservation* toward self-determination
- **High-accuracy Māori language transcription exceeding commercial models** (Community Value): Culturally-Embedded Technology Development *embeds culture* in technically superior outputs
- **Māori cultural governance embedded in AI and archives** (Democratic Practices): Indigenous Knowledge Systems Integration *grounds governance* in tikanga
- **Legal framework protecting Māori data sovereignty** (Democratic Practices): Legal Innovation *formalizes protection* through novel licensing
- **World-leading Māori language AI** (Impact): all conditions combine to produce technological leadership

Synthesis: Te Hiku Media demonstrates how Indigenous epistemological grounding, protective legal innovation, and culturally-embedded technology development combine to produce world-leading language technology under full community control. The case shows that Indigenous technology sovereignty requires not just building good technology but fundamentally restructuring ownership, governance, and purpose—rejecting extractive partnerships, creating novel legal protections, grounding development in Indigenous knowledge systems, and redefining value beyond narrow economic metrics. Te Hiku's Kaitiakitanga License represents a breakthrough in legal innovation that other Indigenous communities are now adopting.

COMPACT EQUATION: For Te Hiku Media, IDS (Partners) + AVD (Problem) + IKSI (Purpose) + CSAKP (Purpose) + LIFCC (Regulation) + CTD (Technology) = ITDR (Bal) + CSAL (Com) + HMLT (Com) + MCGE (Dem) + LFPM (Dem) + WMLA (Imp)

EQUATION 10/43: TE MANA RARAUNGA

Location: New Zealand | **Sector:** Advocacy network - indigenous data rights

Geographic Scope: National, but participate in global networks; Global South

Maturity: 2015; recently more active and seemingly ramping up

Conditions and their mechanisms:

- **Multi-Network Presence** (Partners) *enables coordination* by connecting Māori data sovereignty efforts to global Indigenous movements
- **Policy Influence and Systemic Change** (Politics) *directly produces* formal recognition of Māori data sovereignty in national statistics
- **Decolonial Research Ethics** (Problem) *counters domination* by challenging Western-centric data governance that excludes Indigenous epistemologies
- **Indigenous Knowledge Systems Integration** (Purpose) *grounds governance* by ensuring data governance reflects Māori ways of knowing

Condition interactions: Indigenous Knowledge Systems Integration provides the epistemological foundation that shapes all other work. Decolonial Research Ethics identifies what needs to change in existing frameworks. Multi-Network Presence connects local efforts

to global movements, amplifying voice. Policy Influence and Systemic Change translates these into formal institutional recognition.

Outcomes produced:

- **Transnational Indigenous coordination** (Balance): Multi-Network Presence *enables coordination* amplifying Māori advocacy
- **Formal agreement recognizing Māori data sovereignty** (Impact): Policy Influence *directly produces* institutional change with Stats NZ
- **Culturally grounded governance** (Democratic Practices): Indigenous Knowledge Systems Integration *grounds governance* in Māori values

Synthesis: Te Mana Raraunga demonstrates how Indigenous epistemological grounding, network coordination, and strategic policy engagement combine to produce formal institutional recognition of Māori data sovereignty. The case shows that transforming national data governance requires both conceptual work (establishing what Indigenous data sovereignty means in Māori terms) and strategic engagement (building networks, engaging institutions, securing formal agreements). The Mana Ōrite Agreement with Stats NZ represents a significant achievement in translating Indigenous data sovereignty principles into governmental practice.

COMPACT EQUATION: For Te Mana Raraunga, MP (Partners) + PIASC (Politics) + DRE (Problem) + IKSI (Purpose) = TICA (Bal) + FARM (Imp) + CGGR (Dem) + MDGA (Dem)

EQUATION 11/43: TELECOMUNICACIONES INDÍGENAS COMUNITARIAS (TIC A.C.)

Location: Oaxaca | **Sector:** Indigenous telecommunications provider

Geographic Scope: Local

Maturity: 2016; Mature

Conditions and their mechanisms:

- **Resistance and Construction Integration** (Purpose) *integrates resistance* by combining opposition to regulatory burdens with building functioning infrastructure
- **Prioritizing Local Needs in Global Engagement** (Purpose) *grounds balance* by ensuring technology serves community-defined priorities rather than external agendas
- **Strategic Regulatory Navigation** (Regulation) *navigates regulation* by engaging Mexican spectrum regulation strategically to secure community-serving exemptions
- **Community-Owned Infrastructure** (Technology Tools) *materializes control* by building telecommunications networks that indigenous communities own and operate

Condition interactions: Resistance and Construction Integration provides the strategic orientation—combining opposition with building. Strategic Regulatory Navigation translates this into concrete regulatory victories. Community-Owned Infrastructure provides the material foundation that both motivates regulatory engagement and benefits from regulatory victories. Prioritizing Local Needs ensures that both regulatory strategy and infrastructure development serve community purposes.

Outcomes produced:

- **Legal victory establishing indigenous telecommunications autonomy** (Democratic Practices): Strategic Regulatory Navigation *navigates regulation* to secure exemptions while Community-Owned Infrastructure *materializes* the infrastructure that victories protect
- **Community assemblies determining telecommunications priorities** (Balance): Prioritizing Local Needs *grounds balance* in community decision-making
- **Regulatory exemption enabling sustainable community telecommunications** (Impact): Strategic Regulatory Navigation *navigates regulation* producing lasting policy change

Synthesis: TIC A.C. demonstrates how resistance-construction integration, strategic regulatory navigation, and community infrastructure ownership combine to produce indigenous telecommunications autonomy in hostile regulatory environments. The case shows that community telecommunications in contexts of regulatory hostility requires simultaneous work on multiple fronts: building actual infrastructure that serves communities, engaging regulatory systems strategically to secure operating space, and maintaining community governance over both infrastructure and regulatory engagement. TIC A.C.'s victory—overturning \$50,000 annual spectrum fees—represents a breakthrough that creates space for other community networks across Mexico.

COMPACT EQUATION: For Telecomunicaciones Indígenas Comunitaria, RACI (Purpose) + PLNIGE (Purpose) + SRN (Regulation) + CI (Technology) = LVEI (Dem) + CADT (Bal) + REES (Imp)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 12/43: WORKER ALGORITHM OBSERVATORY (WAO)

Location: USA | **Sector:** Algorithmic justice group

Geographic Scope: National; they tap into global tech worker movements

Maturity: Est. 2022; emerging

Conditions and their mechanisms:

- **Inclusive Expertise Integration** (Governance) *integrates expertise* by combining direct worker involvement with specialized analytical capacity
- **Worker-Led Transparency Mechanisms** (Problem) *exposes systems* by creating mechanisms for workers to document and challenge algorithmic opacity
- **Resistance and Construction Integration** (Purpose) *integrates resistance* by building tools for understanding algorithms alongside advocacy for change
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating worker authority over data about algorithmic management
- **Algorithmic Accountability and Transparency** (Technology Tools) *exposes systems* by building infrastructure that makes opaque algorithmic processes visible

Condition interactions: Worker-Led Transparency Mechanisms and Algorithmic Accountability and Transparency work together to expose what platforms hide. Democratic Data Governance ensures workers control the data they generate through this exposure. Inclusive Expertise Integration enables productive collaboration between workers and researchers. Resistance and Construction Integration ensures that exposure leads to both advocacy and tool-building.

Outcomes produced:

- **Hybrid governance combining worker involvement with expert analysis** (Community Value): Inclusive Expertise Integration *integrates expertise* democratically
- **Worker-controlled tools exposing algorithmic manipulation** (Democratic Practices): Algorithmic Accountability and Transparency *exposes systems* under worker control
- **Worker-generated data enabling advocacy** (Democratic Practices): Democratic Data Governance *establishes control* over worker-generated evidence

Synthesis: WAO demonstrates how worker-led transparency, hybrid expertise integration, and resistance-construction integration combine to produce tools that expose platform algorithmic manipulation while building worker capacity for ongoing advocacy. The case shows that algorithmic accountability requires not just technical analysis but worker-controlled investigation processes that generate evidence workers can use for advocacy. WAO's approach creates ongoing capacity for workers to document, understand, and challenge algorithmic management—addressing the fundamental information asymmetry that enables platform exploitation.

COMPACT EQUATION: For Worker Algorithm Observatory (WAO), IEI (Governance) + WTM (Problem) + RACI (Purpose) + DDG (Technology P) + AAAT (Technology) = HGCD (Com) + WTEA (Dem) + WDEA (Dem) + WAIT (Dem)

Consolidation Notes:

- Merged 1 outcome group(s)

PEER PRODUCTION

12 Organizations

EQUATION 13/43: AUSTRALIAN RESEARCH DATA COMMONS

Location: Australia | **Sector:** Research as a Commons

Geographic Scope: National with some global collaborations

Maturity: Est. 2018; semi-mature

Conditions and their mechanisms:

- **Bidirectional Knowledge Exchange** (Partners) *structures exchange* by creating infrastructure for research data sharing across institutions
- **Technology Design for Multi-Scale Governance** (Technology Practices) *architects federation* by designing infrastructure that supports varied institutional contexts
- **Ethical Data Governance Frameworks** (Regulation) *governs exchange* by creating frameworks that balance openness with protection
- **Public-Community Co-Investment Models** (Resources) *sustains assets* by diversifying funding beyond single sources

Condition interactions: Technology Design for Multi-Scale Governance creates infrastructure that enables Bidirectional Knowledge Exchange at national scale. Ethical Data Governance Frameworks ensures that exchange operates within appropriate protections. Public-Community Co-Investment Models provides the sustainable resourcing that enables ongoing operation.

Outcomes produced:

- **National research data infrastructure** (Balance): Technology Design *architects federation* enabling cross-institutional collaboration
- **Multi-scale data governance** (Balance): Technology Design *designs scale* supporting varied contexts
- **Community-controlled secure data access** (Balance): Ethical Data Governance Frameworks *governs exchange* balancing access and protection
- **Diversified funding for sustainability** (Community Value): Public-Community Co-Investment *sustains assets* long-term

Synthesis: ARDC demonstrates how multi-scale infrastructure design, ethical governance frameworks, and diversified resourcing combine to produce national research data commons that balance openness with protection. The case shows that research data infrastructure requires deliberate design for multiple scales—supporting both institutional autonomy and national coordination—combined with governance frameworks that navigate tensions between openness and protection, and sustainable funding models that prevent dependence on single sources. ARDC's federated approach creates infrastructure that serves varied research contexts while maintaining coherence.

COMPACT EQUATION: For Australian Research Data Commons, BKE (Partners) + TDFMG (Technology P) + EDGF (Regulation) + PCM (Resources) = NRDI (Bal) + MRDG (Bal) + CSDA (Bal) + DFEL (Com)

EQUATION 14/43: GUIFL.NET

Location: Spain | **Sector:** Community wireless internet commons/internet infrastructure

Geographic Scope: National; Global North

Maturity: Est. 2004; mature

Conditions and their mechanisms:

- **Federated Governance Models** (Governance) *configures federation* by establishing governance that coordinates across scales while preserving local autonomy
- **Decentralized Decision-Making** (Org Info) *structures authority* by distributing network governance to local communities
- **Community-Controlled Economic Circulation** (Org Info) *retains value* by keeping telecommunications revenues within local economies
- **Community-Owned Infrastructure Development** (Problem) *motivates practice* by responding to lack of commercial broadband in rural areas
- **Legal Innovation for Community Control** (Regulation) *formalizes protection* by securing legal recognition as public utility commons
- **Economic Transformation and Value Redistribution** (Resources) *transforms flows* by redirecting telecommunications revenues from corporate shareholders to communities
- **Multi-Scalar Federated Governance** (Technology Practices) *architects federation* through technical design enabling both interoperability and local adaptation
- **Community-Owned Infrastructure** (Technology Tools) *materializes control* by creating physical networks communities own

Condition interactions: Community-Owned Infrastructure Development provides the motivating problem. Multi-Scalar Federated Governance and Federated Governance Models work together—technical architecture and governance structure designed for federation. Legal Innovation secures the framework. Economic Transformation ensures value stays local. Decentralized Decision-Making and Community-Controlled Economic Circulation operationalize these at local levels.

Outcomes produced:

- **Comprehensive community-owned telecommunications** (Democratic Practices): Community-Owned Infrastructure *materializes control* at regional scale
- **Local control within shared frameworks** (Democratic Practices): Federated Governance Models *configures federation* preserving autonomy
- **Legal recognition as public utility commons** (Democratic Practices): Legal Innovation *formalizes protection* in novel legal form
- **Federated governance balancing coordination with autonomy** (Balance): Multi-Scalar Federated Governance *architects federation* technically and organizationally
- **Local economic circulation** (Community Value): Community-Controlled Economic Circulation *retains value* locally

- **Economic transformation keeping revenues in communities** (Impact): Economic Transformation *transforms flows* fundamentally

Synthesis: Guifi.net demonstrates how federated governance design, legal innovation, and community ownership combine to produce one of the world's largest community-owned telecommunications networks. The case shows that community infrastructure at scale requires deliberate federation—technical architecture and governance structures designed to enable both local autonomy and network coordination—combined with legal innovation that protects community ownership and economic design that keeps value local. Guifi.net's recognition as a public utility commons represents a legal breakthrough that secures community infrastructure against privatization.

COMPACT EQUATION: For Guifi.net, FGM (Governance) + DD (Org Info) + CEC (Org Info+C57) + CID (Problem) + LIFCC (Regulation) + ETAVR (Resources) + MFG (Technology P) + CI (Technology) = FSCL (Bal) + BFEB (Bal) + LCCO (Dem) + LPMC (Dem) + CCTS (Dem) + LCCO (Imp) + CVTL (Imp) + NCT (Com)

EQUATION 15/43: LINUX FOUNDATION

Location: USA | **Sector:** Software / Technology services

Geographic Scope: Global

Maturity: Est. 1991, foundation in 2007; Mature

Conditions and their mechanisms:

- **Open Source Community Development** (Partners) *structures exchange* by creating governance that balances corporate sponsorship with community control
- **Economic Transformation** (Resources) *sustains commons* by diversifying revenue to maintain organizational independence
- **Open Source Commons Development** (Problem) *motivates practice* by responding to proprietary software concentration
- **Open Access and Transparency** (Technology Practices) *creates transparency* by maintaining open development processes
- **Open-Source Development** (Technology Practices) *constructs commons* by producing software as collectively accessible resource

Condition interactions: Open Source Commons Development provides the orienting problem. Open-Source Development and Open Access and Transparency create the technical commons. Open Source Community Development structures how corporate sponsors and community contributors interact. Economic Transformation ensures organizational sustainability without compromising community control.

Outcomes produced:

- **Balanced engagement with corporate resources** (Balance): Open Source Community Development *structures exchange* preserving community governance
- **Diversified revenue maintaining independence** (Community Value): Economic Transformation *sustains commons* through diversification

- **Collaborative governance ensuring public access** (Community Value): Open Source Community Development *enables coordination* serving public benefit
- **Open collaborative development** (Democratic Practices): Open-Source Development *constructs commons* through transparent processes
- **Essential software accessible and community-driven** (Democratic Practices): Open Access and Transparency *creates transparency* in critical infrastructure

Synthesis: Linux Foundation demonstrates how open source governance, diversified resourcing, and transparent development combine to produce essential software infrastructure that remains accessible while engaging corporate resources. The case shows that sustaining critical software commons requires navigating tensions between corporate engagement and community control—creating governance structures that leverage corporate resources without compromising openness, diversifying funding to maintain independence, and maintaining transparent processes that enable community scrutiny. The result is foundational software infrastructure that serves as genuine public utility.

COMPACT EQUATION: For Linux Foundation, OSCD (Partners) + ET (Resources) + OSCD (Problem) + OAAT (Technology P) + OD (Technology P) = BELC (Bal) + DRSM (Com) + CGEP (Com) + OCDW (Dem) + ESRA (Dem)

EQUATION 16/43: MATRIX

Location: United Kingdom | **Sector:** Internet communications protocol

Geographic Scope: Global

Maturity: Est. 2014; mature

Conditions and their mechanisms:

- **Federated Governance Models** (Governance) *configures federation* by establishing governance for decentralized communication networks
- **Open Standards Development** (Org Info) *defines production* by creating open protocols that enable interoperability
- **Policy Influence and Systemic Change** (Politics) *advocates openness* by engaging policy systems on communications standards
- **Technological Sovereignty Architecture** (Problem) *motivates practice* by responding to tension between local control and global interoperability
- **Open-Source Development** (Technology Practices) *constructs commons* by developing protocols openly
- **Technology Design for Multi-Scale Governance** (Technology Practices) *architects federation* by designing for varied organizational contexts

Condition interactions: Technological Sovereignty Architecture defines the problem: enabling local control without sacrificing interoperability. Federated Governance Models and Technology Design for Multi-Scale Governance address this through both governance structure and technical design. Open Standards Development and Open-Source Development create the protocols and software as commons. Policy Influence engages external systems on behalf of open approaches.

Outcomes produced:

- **Federated architecture combining local sovereignty with global interoperability** (Balance): Technology Design *architects federation* resolving sovereignty-interoperability tension
- **Multi-scale technical architecture** (Balance): Technology Design *designs scale* for varied contexts
- **Community control over communication infrastructure** (Democratic Practices): Federated Governance Models *configures federation* enabling local governance
- **User sovereignty over data and communications** (Democratic Practices): Open Standards Development *defines practice* enabling user control
- **Open protocol shaping global secure communications** (Impact): Policy Influence *advocates openness* achieving international influence
- **Open protocol enabling community control** (Community Value): Open-Source Development *constructs commons* as infrastructure

Synthesis: Matrix demonstrates how federated design, open standards development, and policy engagement combine to produce communication infrastructure that resolves the tension between local sovereignty and global interoperability. The case shows that building genuinely decentralized communications requires technical architecture designed for federation, governance structures that enable local control, open protocols that prevent proprietary lock-in, and policy engagement that shapes broader communications standards. Matrix's growing adoption by governments and institutions demonstrates that federated, open approaches can compete with centralized platforms.

COMPACT EQUATION: For Matrix, FGM (Governance) + OSD (Org Info) + PIASC (Politics) + TSA (Problem) + OD (Technology P) + TDFMG (Technology P) = FACL (Bal) + MTAA (Bal) + CCOC (Dem) + USOP (Dem) + OPSI (Imp) + OPEC (Com)

EQUATION 17/43: NEA GUINEA/TZOUMAKERS

Location: Greece | **Sector:** Maker space for developing sustainability tools, skills, and production systems

Geographic Scope: Local

Maturity: Est. 2009; Mature

Conditions and their mechanisms:

- **Decentralized Decision-Making** (Governance) *structures authority* by distributing design and production decisions to community members
- **Culturally-Embedded Technology Development** (Technology Tools) *embeds culture* by integrating traditional agricultural knowledge with modern fabrication techniques

Condition interactions: Decentralized Decision-Making creates the governance context within which Culturally-Embedded Technology Development can operate—community members have authority to shape what gets made and how. Together, they enable participatory design that draws on local knowledge.

Outcomes produced:

- **Participatory maker space for sustainability solutions** (Democratic Practices): Decentralized Decision-Making *structures authority* in co-design processes
- **Digital tools integrating traditional and modern knowledge** (Balance): Culturally-Embedded Technology Development *embeds culture* in hybrid tools

Synthesis: Nea Guinea/Tzoumakers demonstrates how decentralized governance and culturally-embedded technology development combine to produce a participatory maker space that bridges traditional agricultural knowledge with modern fabrication. The case shows that community-led technology development in traditional contexts requires governance structures that distribute decision-making authority and design processes that genuinely integrate local knowledge rather than imposing external solutions. The result is technology that serves local needs while building on cultural heritage.

COMPACT EQUATION: For Nea Guinea/Tzoumakers, DD (Governance) + CTD (Technology) = PMSE (Dem) + DTIT (Bal)

EQUATION 18/43: OPENAGRI.NET

Location: India | **Sector:** Digital technology for agriculture / agritech

Geographic Scope: Global; Global South

Maturity: Est. 2024; emerging

Conditions and their mechanisms:

- **Decentralized Technological Architecture** (Regulation) *asserts sovereignty* by creating agricultural data infrastructure independent of state and corporate frameworks
- **Technology Design for Multi-Scale Governance** (Technology Practices) *architects federation* by designing for both smallholder farmer governance and global interoperability

Condition interactions: Technology Design for Multi-Scale Governance addresses the technical challenge of serving smallholder farmers while maintaining global interoperability. Decentralized Technological Architecture ensures that farmer governance isn't subordinated to state or corporate frameworks.

Outcomes produced:

- **Open agricultural infrastructure serving smallholders** (Balance): Technology Design *architects federation* for farmer governance
- **Farmer-controlled technology with global interoperability** (Balance): Decentralized Technological Architecture *asserts sovereignty* while enabling connection

Synthesis: OpenAgriNet demonstrates how decentralized architecture and multi-scale design combine to produce agricultural data infrastructure that serves smallholder farmers while

maintaining global connectivity. The case shows that agricultural technology serving small farmers requires deliberate design for their governance needs—not scaling down corporate solutions but building from farmer requirements up—while maintaining interoperability that enables participation in broader markets and knowledge networks.

COMPACT EQUATION: For OpenAgriNet, DTA (Regulation) + TDFMG (Technology P)
= OAIS (Bal) + FATW (Bal)

EQUATION 19/43: OPENSTREETMAP

Location: United Kingdom | **Sector:** Open source maps (maps as a commons)

Geographic Scope: Global

Maturity: 2004; mature - global reach

Conditions and their mechanisms:

- **Global-Local Network Formation** (Social Context) *embeds standards* by situating mapping within both global technical communities and local mapping groups
- **Commons-Based Resource Governance** (Technology Practices) *stewards commons* by governing shared mapping data across diverse communities
- **Open Access and Transparency** (Technology Practices) *creates transparency* by maintaining open access to all geographic data
- **Digital Resources as Commons** (Technology Tools) *instantiates commons* by producing mapping data as freely accessible resource
- **Open Source Commons Development** (Technology Tools) *enables access* by minimizing operational costs through volunteer contributions

Condition interactions: Global-Local Network Formation structures how global standards and local mapping practices interact. Commons-Based Resource Governance and Open Access and Transparency govern the shared resource. Digital Resources as Commons and Open Source Commons Development create and maintain the mapping data as commons.

Outcomes produced:

- **Local mapping autonomy within global commons** (Balance): Global-Local Network Formation *embeds standards* enabling both specificity and universality
- **Shared protocols with community-specific practices** (Balance): Commons-Based Resource Governance *stewards commons* accommodating diversity
- **Open mapping commons accessible globally** (Democratic Practices): Open Access and Transparency *creates transparency* for all users
- **Volunteer-based operations enabling accessibility** (Democratic Practices): Open Source Commons Development *enables access* through distributed contribution
- **Free geographic data enabling local initiatives** (Community Value): Digital Resources as Commons *instantiates commons* with practical utility

Synthesis: OpenStreetMap demonstrates how global-local network formation, commons governance, and open access combine to produce the world's most comprehensive open geographic database. The case shows that global commons require deliberate navigation

between universal standards and local specificity—creating governance that enables both consistency and adaptation, technical infrastructure that supports diverse contributors, and access frameworks that ensure universal availability. OSM's success demonstrates that volunteer-driven commons can compete with and exceed corporate alternatives.

COMPACT EQUATION: For OpenStreetMap, GNF (Social Conte) + CRG (Technology P) + OAAT (Technology P) + DRAC (Technology) + OSCD (Technology) = LMAW (Bal) + SPWC (Bal) + OMCA (Dem) + VOEM (Dem) + FGDE (Com)

EQUATION 20/43: P2P FOUNDATION

Location: Netherlands | **Sector:** Advocacy group - commons-based society
Geographic Scope: Global
Maturity: Est. 2007; Mature

Conditions and their mechanisms:

- **Commons-Based Knowledge Production** (Org Info) *defines production* by orienting organization toward knowledge commons rather than proprietary outputs
- **Bidirectional Knowledge Exchange** (Partners) *circulates knowledge* by creating flows between global and local commons initiatives
- **Multi-Network Presence** (Partners) *enables coordination* by positioning across policy, research, and movement networks
- **Policy Influence and Systemic Change** (Politics) *advocates commons* by engaging policy discourse on commons-based alternatives
- **Digital Resources as Commons** (Technology Tools) *instantiates commons* by producing and maintaining digital knowledge resources

Condition interactions: Commons-Based Knowledge Production orients all other work. Multi-Network Presence and Bidirectional Knowledge Exchange create infrastructure for knowledge circulation. Policy Influence and Systemic Change translates commons advocacy into policy discourse. Digital Resources as Commons creates the practical outputs.

Outcomes produced:

- **Community-maintained global knowledge database** (Community Value): Digital Resources as Commons *instantiates commons* as searchable resource
- **Comprehensive commons ecosystem** (Community Value): Commons-Based Knowledge Production *defines production* across domains
- **Reciprocal knowledge flows** (Balance): Bidirectional Knowledge Exchange *circulates knowledge* between scales
- **Transnational knowledge infrastructure** (Balance): Multi-Network Presence *enables coordination* across commons movement
- **Commons frameworks entering policy discussions** (Impact): Policy Influence *advocates commons* in mainstream discourse

Synthesis: P2P Foundation demonstrates how commons-oriented knowledge production, multi-network positioning, and policy engagement combine to produce both practical

knowledge resources and growing policy influence for commons-based approaches. The case shows that advancing the commons paradigm requires comprehensive infrastructure—documentation of existing commons initiatives, conceptual frameworks for understanding them, network connections across the movement, and policy engagement that introduces commons into mainstream discourse. P2P Foundation's work has helped establish commons as a recognized policy paradigm.

COMPACT EQUATION: For P2P Foundation, CKP (Org Info) + BKE (Partners) + MP (Partners) + PIASC (Politics) + DRAC (Technology) = CGMD (Com) + CCES (Com) + RKFB (Bal) + TKIS (Bal) + CFEP (Imp)

EQUATION 21/43: RESEARCH DATA ALLIANCE

Location: United Kingdom | **Sector:** Community driven data infrastructure for sharing
Geographic Scope: Global
Maturity: Est. 2013; mature

Conditions and their mechanisms:

- **Bidirectional Knowledge Exchange** (Partners) *structures exchange* by creating infrastructure for research data sharing across disciplines and geographies
- **Open Standards Development** (Technology Practices) *adapts standards* by developing flexible frameworks that accommodate disciplinary diversity

Condition interactions: Bidirectional Knowledge Exchange creates the social infrastructure for data sharing. Open Standards Development creates the technical standards that enable sharing across diverse research contexts.

Outcomes produced:

- **Global data sharing infrastructure** (Balance): Bidirectional Knowledge Exchange *structures exchange* at global scale
- **Flexible standards accommodating diverse contexts** (Community Value): Open Standards Development *adapts standards* for disciplinary variation

Synthesis: Research Data Alliance demonstrates how bidirectional exchange and flexible standards development combine to produce global research data sharing infrastructure. The case shows that research data commons require both social infrastructure (relationships and agreements across institutions and disciplines) and technical standards (interoperability frameworks)—with the key insight that standards must be flexible enough to accommodate disciplinary diversity rather than imposing uniformity.

EQUATION 22/43: SENSORICA

Location: Canada | **Sector:** Crowd-sourced open science for fourth industrial revolution tech
Geographic Scope: Global

Maturity: Est. 2010; mature

Conditions and their mechanisms:

- **Stigmergic Coordination Systems** (Governance) *structures governance* through indirect coordination enabling distributed participation without hierarchy
- **Multi-Sector Innovation Networks** (Partners) *connects networks* by bridging corporate, academic, and community actors in collaborative innovation
- **Alternative Economic Models** (Resources) *experiments models* by testing open value network approaches to economic organization
- **Open-Source Development** (Technology Practices) *constructs commons* by developing hardware and knowledge openly

Condition interactions: Stigmergic Coordination Systems enables coordination at scale without hierarchy—participants contribute based on signals rather than direction. Multi-Sector Innovation Networks brings diverse actors into this coordination. Alternative Economic Models experiments with how to distribute value in open networks. Open-Source Development ensures outputs remain commons.

Outcomes produced:

- **Distributed decision-making with transparent contribution tracking** (Democratic Practices): Stigmergic Coordination Systems *structures governance* enabling global peer production
- **Reduced R&D costs through crowd-sourced innovation** (Balance): Multi-Sector Innovation Networks *connects networks* for collaborative efficiency
- **Novel organizational structure for commons-based peer production** (Community Value): Alternative Economic Models *experiments models* creating new forms
- **Open collaborative research enabling distributed innovation** (Community Value): Open-Source Development *constructs commons* through open hardware

Synthesis: Sensorica demonstrates how stigmergic coordination, multi-sector networks, and alternative economic experimentation combine to produce a novel model for open hardware development and commons-based peer production. The case shows that enabling distributed innovation at scale requires coordination mechanisms beyond both hierarchy and market—stigmergic systems that enable contribution based on signals rather than direction, combined with value accounting systems that can fairly distribute rewards across distributed contributors. Sensorica's open value network model represents an important experiment in post-capitalist economic organization.

COMPACT EQUATION: For Sensorica, SCS (Governance) + MIN (Partners) + AEM (Resources) + OD (Technology P) = DDWT (Dem) + RRCT (Bal) + NOSE (Com) + OCRE (Com)

EQUATION 23/43: SOLAR COMMONS

Location: USA | **Sector:** Solar energy as a commons - climate tech
Geographic Scope: Local (Duluth and Tucson)

Maturity: Est. 2017; emerging

Conditions and their mechanisms:

- **Community Asset Development Through Legal Innovation** (Purpose) *directs value* by orienting solar development toward community wealth rather than investor returns
- **Legal Innovation for Community Control** (Regulation) *formalizes protection* by creating trust structures that secure community ownership of solar assets
- **Economic Transformation and Value Redistribution** (Resources) *transforms flows* by redirecting renewable energy benefits from external investors to local communities

Condition interactions: Community Asset Development Through Legal Innovation provides the orienting purpose—framing solar as community wealth opportunity. Legal Innovation for Community Control creates the trust structures that make this possible. Economic Transformation and Value Redistribution operationalizes value retention.

Outcomes produced:

- **Integrated framework connecting energy to community transformation** (Community Value): Community Asset Development *directs value* comprehensively
- **Solar benefits generating long-term community wealth** (Community Value): Economic Transformation *transforms flows* toward communities
- **Community trusts holding solar assets** (Democratic Practices): Legal Innovation *formalizes protection* through trust structures

Synthesis: Solar Commons demonstrates how purposeful community asset development, legal innovation, and value redistribution combine to produce solar infrastructure that generates lasting community wealth rather than extractive investor returns. The case shows that community benefit from renewable energy requires deliberate legal and economic design—not just installing solar panels but structuring ownership, governance, and value flows so that benefits accrue to communities long-term. Solar Commons' trust model represents an innovation that other communities are now adapting.

COMPACT EQUATION: For Solar Commons, CADTLI (Purpose) + LIFCC (Regulation) + ETAVR (Resources) = IFCE (Com) + SBGL (Com) + CTHS (Dem)

EQUATION 24/43: WIKIHOUSE

Location: United Kingdom | **Sector:** Open source digital fabrication system for modular house construction

Geographic Scope: Global

Maturity: Est. 2011; mature

Conditions and their mechanisms:

- **Open Source Manufacturing Networks** (Org Info) *positions* WikiHouse as infrastructure for distributed housing production
- **Open-Source Development** (Technology Practices) *constructs commons* by developing housing designs that anyone can access, modify, and produce
- **Digital Resources as Commons** (Technology Tools) *instantiates commons* by creating freely accessible construction knowledge

Condition interactions: Open Source Manufacturing Networks provides the organizational model. Open-Source Development creates the designs as commons. Digital Resources as Commons ensures practical accessibility.

Outcomes produced:

- **Open-source construction enabling local fabrication** (Community Value): Open-Source Development *constructs commons* for local production
- **Freely accessible housing designs** (Community Value): Digital Resources as Commons *instantiates commons* as practical resource
- **Housing knowledge commons for local fabrication** (Community Value): all conditions combine to democratize construction knowledge

Synthesis: WikiHouse demonstrates how open source organizational models and commons-based production combine to produce housing designs that democratize construction knowledge. The case shows that extending open source principles to physical production requires not just releasing designs but creating infrastructure for distributed fabrication—design systems that enable local adaptation and production, documentation that supports non-expert builders, and licensing that prevents enclosure. WikiHouse represents an important experiment in extending digital commons principles to the built environment.

COMPACT EQUATION: For WikiHouse, OSMN (Org Info) + OD (Technology P) + DRAC (Technology) = OCEL (Com) + FAHD (Com) + HKCA (Com)

DIGITAL COOPERATIVE

11 Organizations

EQUATION 25/43: COOPCYCLE

Location: France | **Sector:** Federation of logistics messengers / cooperatives

Geographic Scope: Regional (Europe); Global North

Maturity: Est. 2016; mature

Conditions and their mechanisms:

- **Worker Ownership Structures** (Governance) *retains value* by anchoring logistics work in cooperative ownership
- **Scale and Democratic Participation Navigation** (Governance) *structures governance* to maintain democratic participation as the federation grows
- **Federated Scaling Strategy** (Governance) *configures federation* by balancing growth with individual cooperative autonomy
- **Legal Innovation for Community Control** (Regulation) *formalizes protection* by creating licensing that prevents corporate appropriation of cooperative technology
- **Value Redistribution Mechanisms** (Resources) *redistributes value* by keeping delivery earnings with workers rather than platform shareholders

Condition interactions: Worker Ownership Structures provides the foundational alternative to gig platforms. Federated Scaling Strategy and Scale and Democratic Participation Navigation address how to grow while maintaining democratic character. Legal Innovation for Community Control protects cooperative technology from corporate capture. Value Redistribution Mechanisms operationalizes the economic alternative.

Outcomes produced:

- **Worker-owned alternative to exploitative platforms** (Community Value): Worker Ownership Structures *retains value* for workers
- **Worker-retained value through cooperative ownership** (Community Value): Value Redistribution Mechanisms *redistributes value* to members
- **Federated structure balancing efficiency with participatory governance** (Democratic Practices): Federated Scaling Strategy *configures federation* preserving democracy
- **Legal protection ensuring technology serves cooperative movement** (Democratic Practices): Legal Innovation *formalizes protection* against appropriation
- **Transnational growth with locally-responsive governance** (Impact): Federated Scaling Strategy enables *scale* while preserving local autonomy

Synthesis: CoopCycle demonstrates how worker ownership, federated governance, and legal innovation combine to produce a transnational delivery platform that retains value for workers while maintaining democratic governance at scale. The case shows that platform cooperativism requires addressing multiple dimensions simultaneously: ownership structures that anchor worker control, federation strategies that enable growth without sacrificing local autonomy, and legal innovations that prevent corporate appropriation of cooperative

technology. CoopCycle's Coopyleft license represents an important innovation for protecting cooperative technology commons.

COMPACT EQUATION: For CoopCycle, WOS (Governance) + SADPN (Governance) + FSS (Governance) + LIFCC (Regulation) + VRM (Resources) = WATE (Com) + WVTC (Com) + FSBE (Dem) + LPET (Dem) + TGWL (Imp)

EQUATION 26/43: FAIRBNB.COOP

Location: Italy | **Sector:** Community-centred vacation rental platform (tourism booking)

Geographic Scope: Europe

Maturity: Est. 2014; semi-mature

Conditions and their mechanisms:

- **Anti-Extractive Value Distribution** (Org Info) *defines boundaries* by structuring the platform to reinvest in communities rather than extract to shareholders
- **Community-Controlled Technology Development** (Partners) *protects sovereignty* by maintaining cooperative governance over technology choices
- **Economic Transformation and Value Redistribution** (Resources) *transforms flows* by directing platform fees to community projects

Condition interactions: Anti-Extractive Value Distribution provides the foundational orientation. Community-Controlled Technology Development ensures technology choices serve cooperative values. Economic Transformation and Value Redistribution operationalizes community benefit.

Outcomes produced:

- **Platform fees reinvested in community projects** (Community Value): Economic Transformation *transforms flows* to local benefit
- **Technology aligned with cooperative values** (Balance): Community-Controlled Technology Development *protects sovereignty* over technical choices
- **Alternative platform demonstrating non-extractive model** (Community Value): Anti-Extractive Value Distribution *defines* the alternative

Synthesis: Fairbnb.coop demonstrates how anti-extractive organizational design, community-controlled development, and value redistribution combine to produce a vacation rental platform that benefits rather than harms host communities. The case shows that platform cooperativism in the sharing economy requires fundamental redesign of value flows—not just worker ownership but community benefit structures that address the externalities (housing displacement, over-tourism) that extractive platforms create.

COMPACT EQUATION: For Fairbnb.coop, AVD (Org Info) + CSWGN (Partners) + VRM (Resources) = CPFT (Com) + TRRT (Com) + LGMC (Bal)

EQUATION 27/43: GROUPMUSE

Location: USA | **Sector:** Community music events platform

Geographic Scope: USA but slowly expanding - music is transnational; Global North

Maturity: Est. 2013, coop conversion in 2020; reasonably mature

Conditions and their mechanisms:

- **Community Ownership for Wealth Retention** (Governance) *retains value* by anchoring cultural production in cooperative ownership
- **Value Reconceptualization** (Purpose) *redefines value* by prioritizing cultural and social value over profit extraction

Condition interactions: Value Reconceptualization provides the foundational reframing—music events as community value creation rather than commercial transaction. Community Ownership for Wealth Retention structures ownership to retain that value.

Outcomes produced:

- **Cooperative structure retaining cultural value** (Community Value): Community Ownership *retains value* within community
- **Community music events creating cultural and social value** (Community Value): Value Reconceptualization *redefines value* as cultural

Synthesis: Groupmuse demonstrates how cooperative ownership and value reconceptualization combine to produce a platform for house concerts that retains cultural value within communities rather than extracting it. The case shows that cooperative alternatives in cultural sectors require not just ownership restructuring but fundamental reframing of what counts as value—centering cultural and social benefit rather than profit metrics.

COMPACT EQUATION: For Groupmuse, COFWR (Governance) + VR (Purpose) = CSRC (Com) + CMEC (Com)

EQUATION 28/43: LES MERCEDES

Location: Spain | **Sector:** Delivery service

Geographic Scope: Local

Maturity: Est. 2020; emerging

Conditions and their mechanisms:

- **Feminist Economic Transformation** (Purpose) *orients practice* by centering gender equity in delivery work organization
- **Prioritizing Local Needs in Global Engagement** (Org Info) *positions scale* by grounding alternatives in specific neighborhood contexts

- **Alternative Economic Models (Resources)** *experiments models* by testing cooperative approaches to delivery work

Condition interactions: Feminist Economic Transformation provides the distinctive orientation—addressing patriarchal exclusion in urban mobility work. Prioritizing Local Needs ensures alternatives serve specific neighborhood contexts. Alternative Economic Models enables experimentation with cooperative forms.

Outcomes produced:

- **Inclusive employment with democratic workplace policies** (Democratic Practices): Feminist Economic Transformation *orients practice* toward gender equity
- **Locally-rooted alternative addressing neighborhood needs** (Balance): Prioritizing Local Needs *positions scale* locally
- **Cooperative structure providing fair wages and protections** (Community Value): Alternative Economic Models *experiments* cooperative forms

Synthesis: Les Mercedes demonstrates how feminist economic orientation, local grounding, and cooperative experimentation combine to produce delivery work alternatives that center gender equity and neighborhood needs. The case shows that platform cooperativism can address multiple dimensions of exclusion simultaneously—not just worker exploitation but gendered exclusion from urban mobility work—while remaining grounded in specific local contexts rather than scaling generic solutions.

COMPACT EQUATION: For Les Mercedes, FET (Purpose) + PLNIGE (Org Info) + AEM (Resources) = IEOW (Dem) + LAAS (Bal) + CSPF (Com)

EQUATION 29/43: LOOMIO

Location: New Zealand | **Sector:** Software / Technology services

Geographic Scope: Global

Maturity: Established 2011; mature, global technology services firm

Conditions and their mechanisms:

- **Decentralized Decision-Making** (Governance) *structures authority* by distributing organizational governance to all cooperative members
- **Collective Ownership and Governance** (Org Info) *defines ownership* as collective worker ownership of the software enterprise
- **Open-Source Development** (Technology Practices) *constructs commons* by developing decision-making tools as open source
- **Transparent and Participatory Design** (Technology Tools) *operationalizes transparency* by building tools that enable inclusive deliberation

Condition interactions: Collective Ownership and Governance and Decentralized Decision-Making together structure internal democratic practice. Open-Source Development ensures the tools themselves are commons. Transparent and Participatory Design creates technology that enables deliberation across barriers.

Outcomes produced:

- **Non-hierarchical cooperative governance** (Democratic Practices): Decentralized Decision-Making *structures authority* without hierarchy
- **Rotating coordination with deliberation-based decisions** (Democratic Practices): Collective Ownership *defines ownership* enabling rotation
- **Asynchronous participation enabling inclusive decisions** (Democratic Practices): Transparent and Participatory Design *operationalizes* accessibility
- **Free and open tools for democratic deliberation** (Community Value): Open-Source Development *constructs commons* as decision infrastructure

Synthesis: Loomio demonstrates how worker cooperative governance, open source development, and participatory design combine to produce both a democratically-governed organization and tools that enable democratic deliberation in other organizations. The case shows that creating technology for democracy requires practicing democracy—Loomio's internal cooperative governance informs the design of tools that enable deliberation, creating a feedback loop between organizational practice and technology development.

COMPACT EQUATION: For Loomio, DD (Governance) + COAG (Org Info) + OD (Technology P) + TAPD (Technology) = NCGW (Dem) + RCRW (Dem) + APEI (Dem) + FAOT (Com)

EQUATION 30/43: MiDATA

Location: Switzerland | **Sector:** Data cooperative for health data

Geographic Scope: National; but permits global projects

Maturity: Est. 2015; semi mature

Conditions and their mechanisms:

- **Community Ownership for Wealth Retention** (Governance) *retains value* by anchoring health data control in cooperative member ownership
- **Federated Democratic Governance** (Governance) *configures federation* by enabling cross-border coordination while preserving local data sovereignty
- **Cooperative Data Stewardship** (Org Info) *defines ownership* as collective member control over health data
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating technical mechanisms for individual consent and collective governance

Condition interactions: Cooperative Data Stewardship and Community Ownership for Wealth Retention together anchor data governance in member control. Federated Democratic Governance enables coordination across cooperatives. Democratic Data Governance provides the technical mechanisms for consent and governance.

Outcomes produced:

- **Cooperative data governance retaining benefits for members** (Community Value): Community Ownership *retains value* in member hands

- **Citizen-controlled health data with consent management** (Democratic Practices): Democratic Data Governance *establishes control* through consent mechanisms
- **Interconnected cooperatives enabling research while preserving control** (Balance): Federated Democratic Governance *configures federation* for coordination

Synthesis: MiData demonstrates how cooperative ownership, federated governance, and democratic data practices combine to produce health data infrastructure that enables beneficial research while maintaining individual and collective control. The case shows that data cooperativism requires not just ownership restructuring but governance design that enables both individual consent and collective decision-making, combined with federation that enables coordination across contexts without sacrificing local sovereignty.

COMPACT EQUATION: For MiData, COFWR (Governance) + FDG (Governance) + CDS (Org Info) + DDG (Technology P) = CDGR (Com) + CHDS (Dem) + ICEC (Bal)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 31/43: NAMMA YATRI

Location: India | **Sector:** Ride-hailing platform - mobility / transportation

Geographic Scope: Local (Bangalore); Global South

Maturity: Est. 2022; emerging

Conditions and their mechanisms:

- **Worker-Owned Platform Alternative** (Org Info) *defines ownership* as driver ownership of the ride-hailing platform
- **Worker-Owned Platform Alternative** (Politics) *navigates balance* between market service provision and transformative non-extractive values
- **Economic Transformation and Value Redistribution** (Resources) *transforms flows* by eliminating commission extraction from drivers

Condition interactions: Worker-Owned Platform Alternative (Org Info) establishes the ownership foundation. Worker-Owned Platform Alternative (Politics) navigates the strategic challenge of market engagement. Economic Transformation operationalizes zero-commission model.

Outcomes produced:

- **Worker ownership with full earnings retention** (Community Value): Worker-Owned Platform Alternative *defines ownership* with Economic Transformation *eliminating* extraction
- **Market-engaged alternative demonstrating viability** (Impact): Worker-Owned Platform Alternative (Politics) *navigates balance* proving non-extractive models work
- **Zero-commission model maximizing driver earnings** (Impact): Economic Transformation *transforms flows* fundamentally

Synthesis: Namma Yatri demonstrates how worker ownership and economic transformation combine to produce a zero-commission ride-hailing platform that proves non-extractive models can compete in the market. The case shows that platform cooperativism can directly challenge extractive platforms by eliminating the commission that enables extraction—creating proof of concept that worker-owned alternatives are economically viable while dramatically improving worker earnings.

COMPACT EQUATION: For Namma Yatri, WPA (Org Info) + WPA (Politics) + ETAVR (Resources) = WOWF (Com) + MADV (Imp) + ZMMD (Imp)

EQUATION 32/43: SOLSHARE

Location: Bangladesh | **Sector:** Climate Technology / Energy Marketplace

Geographic Scope: (inter)National, though some products are sold in India and their tech partner is German; Global Sou

Maturity: Est. 2014; diversified business lines, reasonably mature

Conditions and their mechanisms:

- **Climate Crisis Response** (Social Context) *responds crisis* by grounding energy innovation in climate vulnerability and infrastructure failure
- **Technology Design for Multi-Scale Governance** (Technology Practices) *architects federation* by designing peer-to-peer energy systems that enable local trading within broader grids
- **Community-Owned Infrastructure** (Technology Tools) *materializes control* by creating energy infrastructure communities can own and operate
- **Decentralized Technological Architecture** (Technology Tools) *decentralizes technically* by enabling peer-to-peer electricity trading

Condition interactions: Climate Crisis Response provides urgency and orientation. Technology Design for Multi-Scale Governance and Decentralized Technological Architecture together create the technical architecture for peer-to-peer trading. Community-Owned Infrastructure ensures communities own the resulting systems.

Outcomes produced:

- **Community-controlled energy systems providing climate resilience** (Impact): Climate Crisis Response *responds crisis* through community energy
- **Multi-scale energy infrastructure** (Balance): Technology Design *architects federation* enabling local and grid integration
- **Community energy sharing maximizing solar utilization** (Democratic Practices): Community-Owned Infrastructure *materializes control* over energy
- **Real-time peer-to-peer electricity trading** (Community Value): Decentralized Technological Architecture *decentralizes* energy trading

Synthesis: SOLshare demonstrates how climate crisis response, multi-scale architecture, and decentralized technology combine to produce peer-to-peer energy trading that maximizes solar utilization while building community climate resilience. The case shows that

community energy innovation requires technical architecture designed for peer-to-peer trading within broader grid contexts—not off-grid isolation but integration that maintains community control while enabling efficient resource sharing.

COMPACT EQUATION: For SOLshare, CCR (Social Conte) + TDFMG (Technology P) + CI (Technology) + DTA (Technology) = CESP (Imp) + MEIE (Bal) + CESI (Dem) + RPET (Com)

EQUATION 33/43: SOM ENERGIA

Location: Spain | **Sector:** Renewable energy

Geographic Scope: National; they participate in European renewable energy networks

Maturity: Est. 2010; mature

Conditions and their mechanisms:

- **Community Ownership for Wealth Retention** (Governance) *retains value* by anchoring renewable energy in cooperative member ownership
- **Community Resilience and Ecological Sustainability** (Purpose) *orients practice* toward integration of economic and ecological transformation

Condition interactions: Community Ownership for Wealth Retention structures economic benefit retention. Community Resilience and Ecological Sustainability ensures economic transformation serves broader ecological and social purposes.

Outcomes produced:

- **Cooperative renewable energy keeping benefits in community** (Community Value): Community Ownership *retains value* for members
- **Integrated measurement capturing economic and social dimensions** (Impact): Community Resilience and Ecological Sustainability *orients* comprehensive transformation

Synthesis: Som Energia demonstrates how cooperative ownership and ecological purpose combine to produce renewable energy infrastructure that keeps benefits within member communities while advancing broader energy transition. The case shows that community energy cooperatives can achieve scale—Som Energia has over 70,000 members—while maintaining democratic governance and ecological orientation.

COMPACT EQUATION: For Som Energia, COFWR (Governance) + CRAES (Purpose) = CREK (Com) + IMCE (Imp)

EQUATION 34/43: THE DRIVERS COOPERATIVE

Location: USA | **Sector:** Ride-hailing platform - mobility / transportation

Geographic Scope: Local (NYC and Denver)

Maturity: Est. 2020; emerging

Conditions and their mechanisms:

- **Worker Ownership Structures** (Governance) *retains value* by anchoring ride-hailing in driver cooperative ownership
- **Collective Ownership and Governance** (Governance) *structures governance* through democratic driver control over platform decisions
- **Policy Influence and Systemic Change** (Politics) *advocates transformation* by engaging policy discourse on platform labor
- **Economic Transformation and Value Redistribution** (Resources) *transforms flows* by restructuring commission extraction
- **Transparent and Participatory Design** (Technology Practices) *balances* driver privacy with cooperative transparency

Condition interactions: Worker Ownership Structures and Collective Ownership and Governance together anchor driver control. Economic Transformation operationalizes value retention. Policy Influence and Systemic Change translates cooperative success into policy advocacy. Transparent and Participatory Design navigates tensions between transparency and privacy.

Outcomes produced:

- **Driver-members earning 8-10% more per trip** (Community Value): Economic Transformation *transforms flows* increasing earnings
- **Democratic governance with transparent pricing** (Democratic Practices): Collective Ownership *structures governance* democratically
- **Modified transparency respecting privacy** (Democratic Practices): Transparent and Participatory Design *balances* transparency with privacy
- **Worker-owned model informing regulatory reform** (Impact): Policy Influence *advocates transformation* through demonstrated success
- **Transparent worker-controlled platform** (Impact): all conditions combine for platform transformation

Synthesis: The Drivers Cooperative demonstrates how worker ownership, democratic governance, and economic transformation combine to produce a driver-owned ride-hailing platform that proves cooperative alternatives can succeed in competitive markets. The case shows that platform cooperativism requires simultaneous work on ownership, governance, economics, and policy—creating alternatives that demonstrate viability while advocating for regulatory frameworks that support worker ownership. TDC's success in increasing driver earnings by 8-10% provides concrete evidence for policy engagement.

COMPACT EQUATION: For The Drivers Cooperative, WOS (Governance) + COAG (Governance) + PIASC (Politics) + ETAVR (Resources) + TAPD (Technology P) = DEM (Com) + DGWT (Dem) + MTAR (Dem) + WPMI (Imp) + TWPW (Imp)

EQUATION 35/43: ZENZELENI

Location: South Africa | **Sector:** Internet service provider

Geographic Scope: National; Global South

Maturity: Est. 2012; mature

Conditions and their mechanisms:

- **Community Revenue Retention** (Org Info) *retains value* by structuring telecommunications to circulate revenue locally
- **Capacity Building and Technological Empowerment** (Resources) *enables self-governance* by building local technical capacity for network maintenance
- **Community-Owned Infrastructure** (Technology Tools) *materializes control* by creating telecommunications networks communities own

Condition interactions: Community-Owned Infrastructure provides the material foundation. Capacity Building and Technological Empowerment ensures communities can maintain what they own. Community Revenue Retention structures economic benefit retention.

Outcomes produced:

- **Local economic circulation and reinvestment** (Community Value): Community Revenue Retention *retains value* locally
- **Local technical capacity for network maintenance** (Democratic Practices): Capacity Building *enables self-governance* through skills
- **Community-owned telecommunications reducing costs** (Democratic Practices): Community-Owned Infrastructure *materializes control* affordably

Synthesis: Zenzeleni demonstrates how community infrastructure ownership, local capacity building, and revenue retention combine to produce telecommunications that serves rural South African communities while keeping economic benefits local. The case shows that community telecommunications requires not just infrastructure but the capacity to maintain it—building local technical skills alongside physical networks—combined with economic structures that prevent external value extraction.

COMPACT EQUATION: For Zenzeleni, CRR (Org Info) + CBATE (Resources) + CI (Technology) = LECA (Com) + LTCF (Dem) + CTSR (Dem)

PARTICIPATORY GOVERNANCE

8 Organizations

EQUATION 36/43: CLEAR (CIVIC LABORATORY FOR ENVIRONMENTAL ACTION RESEARCH)

Conditions and their mechanisms:

- **Decolonial Research Ethics** (Partners) *integrates knowledge* by restructuring relationships between scientific expertise and community knowledge authority
- **Resistance and Construction Integration** (Politics) *integrates resistance* by combining opposition to harmful systems with building constructive alternatives
- **Policy Influence and Systemic Change** (Politics) *directly produces* policy change through community science engagement
- **Democratizing Access and Skills** (Resources) *democratizes skills* by making specialized research equipment accessible to communities
- **Inclusive Participation and Representation** (Social Context) *enables inclusion* by structuring research participation to center marginalized perspectives

Condition interactions: Decolonial Research Ethics restructures knowledge relationships as the foundation. Democratizing Access and Skills enables community participation in research. Inclusive Participation and Representation ensures marginalized voices are centered. Resistance and Construction Integration maintains strategic orientation. Policy Influence translates research into policy change.

Outcomes produced:

- **Community control over research affecting their environments** (Impact): Decolonial Research Ethics *integrates knowledge* under community authority
- **Community science informing policy change** (Impact): Policy Influence *directly produces* change through community evidence
- **Integrated approach building alternatives while challenging harm** (Democratic Practices): Resistance and Construction Integration *integrates* opposition with construction
- **Citational politics centering marginalized knowledge** (Democratic Practices): Inclusive Participation *enables inclusion* epistemologically
- **Accessible research tools democratizing inquiry** (Community Value): Democratizing Access *democratizes* research capacity

Synthesis: CLEAR demonstrates how decolonial research ethics, equipment democratization, and inclusive participation combine to produce community science that generates both knowledge and policy change while centering marginalized perspectives. The case shows that decolonizing research requires not just acknowledging community knowledge but restructuring research relationships so communities control the process—combined with practical democratization of research tools and deliberate inclusion of marginalized voices in knowledge production.

COMPACT EQUATION: For CLEAR (Civic Laboratory for Environmenta, DRE (Partners) + RACI (Politics) + PIASC (Politics) + DAAS (Resources) + IPAR (Social Conte) = CCOR (Imp) + CSIP (Imp) + IABA (Dem) + CPCM (Dem) + ARTD (Com)

EQUATION 37/43: DECODE

Location: Europe | **Sector:** Demonstration of centering data sovereignty in how we use the internet

Geographic Scope: Regional; Global North

Maturity: Est. 2017; concluded in Dec 2019

Conditions and their mechanisms:

- **Bidirectional Knowledge Exchange** (Partners) *structures exchange* by creating multi-partner ecosystem for citizen data governance
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating mechanisms for citizen authority over their data
- **Open-Source Development** (Technology Practices) *constructs commons* by developing data sovereignty tools as open source

Condition interactions: Bidirectional Knowledge Exchange creates the partnership ecosystem. Democratic Data Governance provides the governance mechanisms. Open-Source Development ensures tools are available as commons for other cities.

Outcomes produced:

- **Multi-partner ecosystem for citizen data governance** (Balance): Bidirectional Knowledge Exchange *structures exchange* across partners
- **Citizen-controlled data governance integrated with municipal democracy** (Democratic Practices): Democratic Data Governance *establishes control* at municipal scale
- **Open-source data sovereignty tools** (Community Value): Open-Source Development *constructs commons* for replication

Synthesis: DECODE demonstrates how multi-partner exchange, democratic data governance, and open source development combine to produce citizen-controlled data infrastructure integrated with municipal democracy. The case shows that municipal data sovereignty requires both technical tools and governance mechanisms—creating the technical infrastructure for citizen data control while integrating with existing democratic institutions—with open source development ensuring tools can be adopted by other cities.

COMPACT EQUATION: For DECODE, BKE (Partners) + DDG (Technology P) + OD (Technology P) = MEEC (Bal) + CDGI (Dem) + ODST (Com)

EQUATION 38/43: ENSPIRAL

Location: New Zealand | **Sector:** Collaboration platform for mutual aid

Geographic Scope: Global
Maturity: Est. 2010; semi-mature

Conditions and their mechanisms:

- **Decentralized Decision-Making** (Governance) *structures authority* by distributing organizational governance across the network
- **Multi-Network Presence** (Partners) *enables coordination* by connecting social enterprises for mutual support
- **Alternative Economic Models** (Resources) *experiments models* by testing participatory resource allocation across enterprises

Condition interactions: Decentralized Decision-Making creates governance structures for distributed authority. Multi-Network Presence connects enterprises. Alternative Economic Models enables economic experimentation through participatory budgeting.

Outcomes produced:

- **Self-managed collaborative structure** (Democratic Practices): Decentralized Decision-Making *structures authority* without hierarchy
- **Collaborative network amplifying impact** (Balance): Multi-Network Presence *enables coordination* for collective benefit
- **Participatory resource allocation** (Community Value): Alternative Economic Models *experiments* with collective budgeting

Synthesis: Enspiral demonstrates how decentralized governance, network coordination, and participatory economics combine to produce a collaborative network of social enterprises that amplify each other's impact. The case shows that supporting social enterprise requires moving beyond isolated organizations to networked forms—creating governance structures for distributed decision-making, connection infrastructure for mutual support, and economic mechanisms for participatory resource allocation.

COMPACT EQUATION: For Enspiral, DD (Governance) + MP (Partners) + AEM (Resources) = SCSD (Dem) + CNAS (Bal) + PRAA (Com)

EQUATION 39/43: OPEN MOBILITY FOUNDATION

Location: United States | **Sector:** Mobility
Geographic Scope: Global; mainly Global North cities
Maturity: Est. 2019; reasonably mature with global reach

Conditions and their mechanisms:

- **Municipal/Public Digital Governance** (Org Info) *positions* OMF as infrastructure for city control over mobility data
- **Multi-Network Presence** (Partners) *enables coordination* by connecting cities for collaborative governance

- **Policy Influence and Systemic Change** (Politics) *directly produces* regulatory frameworks for municipal data authority
- **Strategic Regulatory Navigation** (Regulation) *navigates regulation* to balance municipal autonomy with interoperability
- **Transparent and Participatory Design** (Technology Tools) *operationalizes transparency* through public technology development
- **Technology Design for Multi-Scale Governance** (Technology Practices) *architects federation* for cross-city coordination

Condition interactions: Municipal/Public Digital Governance provides the foundational positioning. Multi-Network Presence and Technology Design for Multi-Scale Governance create infrastructure for city coordination. Policy Influence and Strategic Regulatory Navigation engage external systems. Transparent and Participatory Design ensures accountability.

Outcomes produced:

- **City-managed mobility data with community governance** (Balance): Municipal/Public Digital Governance *positions* cities as data governors
- **Network of cities collaborating on mobility standards** (Balance): Multi-Network Presence *enables coordination* across cities
- **Regulatory frameworks establishing municipal authority** (Impact): Policy Influence *directly produces* regulatory change
- **Public technology development with accountability** (Democratic Practices): Transparent and Participatory Design *operationalizes* public accountability

Synthesis: Open Mobility Foundation demonstrates how municipal positioning, city network coordination, and policy engagement combine to produce public infrastructure for city control over mobility data. The case shows that municipal data sovereignty requires collective action—individual cities lack leverage against global mobility platforms, but networks of cities can establish standards, share development costs, and advocate collectively for regulatory frameworks that recognize municipal data authority.

COMPACT EQUATION: For Open Mobility Foundation, MDIG (Org Info) + MP (Partners) + PIASC (Politics) + SRN (Regulation) + TAPD (Technology) + TDFMG (Technology P) = CMDS (Bal) + NOCC (Bal) + CMGW (Bal) + RFRD (Imp) + PTDW (Dem)

Consolidation Notes:

- Merged 1 outcome group(s)

EQUATION 40/43: OPENSAFELY

Location: United Kingdom | **Sector:** Open source platform for electronic health records

Geographic Scope: National; Global North

Maturity: Est. 2020; emerging

Conditions and their mechanisms:

- **Privacy-Preserving Technology Design** (Partners) *protects sovereignty* by structuring partnerships that balance researcher access with patient data protection
- **Ethical Data Governance Frameworks** (Regulation) *governs exchange* by creating ethical frameworks for health data research
- **Bidirectional Knowledge Exchange** (Resources) *structures exchange* by enabling research while maintaining public accountability
- **Democratic Data Governance** (Technology Practices) *establishes control* by creating transparent mechanisms for health data research

Condition interactions: Privacy-Preserving Technology Design and Ethical Data Governance Frameworks together structure the balance between access and protection. Democratic Data Governance creates transparent mechanisms. Bidirectional Knowledge Exchange enables productive research partnerships.

Outcomes produced:

- **Transparent privacy-preserving research** (Impact): all conditions combine for accountable health data research
- **Context-specific privacy standards** (Balance): Ethical Data Governance Frameworks *governs exchange* with contextual sensitivity
- **Pro bono institutional support** (Balance): Bidirectional Knowledge Exchange *structures exchange* for mutual benefit
- **Privacy-preserving research architecture** (Democratic Practices): Privacy-Preserving Technology Design *protects* while enabling research

Synthesis: OpenSAFELY demonstrates how privacy-preserving design, ethical governance, and transparent data practices combine to produce health data research infrastructure that enables beneficial research while maintaining public accountability. The case shows that privacy-preserving research requires not just technical solutions but ethical frameworks and governance mechanisms—creating comprehensive infrastructure that makes the case for transparency against assumptions that health research requires opacity.

COMPACT EQUATION: For OpenSAFELY, PTD (Partners) + EDGF (Regulation) + BKE (Resources) + DDG (Technology P) = TPRW (Imp) + CPSW (Bal) + PBIS (Bal) + PRAM (Dem)

EQUATION 41/43: PARTICIPATORY BRAZIL

Location: Brazil | **Sector:** Government administration

Geographic Scope: National; Global South

Maturity: Est. March 2014; still emerging

Conditions and their mechanisms:

- **Democratization of Governance and Technology** (Governance) *structures governance* by creating mechanisms for citizen participation in government
- **Decentralized Decision-Making** (Governance) *distributes authority* by enabling citizen input on policy decisions

- **Transparent and Participatory Design** (Technology Tools) *operationalizes transparency* through platforms for citizen engagement
- **Hybrid Democratic Participation Architecture** (Org Info) *positions* the platform as bridge between digital and deliberative democracy
- **Community Capacity Building** (Resources) *builds capacity* for effective citizen policy participation
- **Purpose as Civic Obligation** (Purpose) *orients practice* by framing participation as responsibility not optional engagement
- **Inclusive Participation and Representation** (Social Context) *enables inclusion* by structuring participation to reach excluded citizens

Condition interactions: Democratization of Governance and Technology and Decentralized Decision-Making create the governance framework. Transparent and Participatory Design provides the technical platform. Community Capacity Building ensures citizens can participate effectively. Inclusive Participation and Representation extends reach. Purpose as Civic Obligation provides motivational framing.

Outcomes produced:

- **Government platform enabling citizen participation** (Impact): all governance conditions combine for participatory infrastructure
- **Enhanced citizen capacity for policy participation** (Impact): Community Capacity Building *builds capacity* at scale
- **Online consultation platforms for citizen input** (Democratic Practices): Transparent and Participatory Design *operationalizes* engagement
- **Participatory policy development** (Democratic Practices): Decentralized Decision-Making *distributes authority* to citizens
- **Digital-physical integration** (Democratic Practices): Hybrid Democratic Participation Architecture *positions* platform for comprehensive engagement
- **Citizen co-creation of governance** (Democratic Practices): Purpose as Civic Obligation *orients* participation as responsibility

Synthesis: Participatory Brazil demonstrates how digital platforms, capacity building, and inclusive design combine to produce government infrastructure for citizen participation in policy-making. The case shows that digital participatory governance requires comprehensive infrastructure—not just platforms but capacity building for effective participation, inclusion mechanisms for marginalized citizens, and cultural framing that positions participation as civic responsibility. The integration of digital and deliberative elements addresses limitations of purely online engagement.

COMPACT EQUATION: For Participatory Brazil, DOGAT (Governance) + DD (Governance) + TAPD (Technology) + HDPA (Org Info) + CCB (Resources) + PACO (Purpose) + IPAR (Social Conte) = GPEC (Imp) + ECCF (Imp) + OCPE (Dem) + PPDD (Dem) + DIEC (Dem) + CCOG (Dem) + DPDW (Dem)

EQUATION 42/43: SLUM DWELLERS INTERNATIONAL

Location: South Africa | **Sector:** Slum data for advocacy
Geographic Scope: Global (Global South)

Maturity: 1996; Mature

Conditions and their mechanisms:

- **Democratization of Governance and Technology** (Governance) *structures governance* by enabling resident-led data collection and advocacy
- **Prioritizing Local Needs in Global Engagement** (Org Info) *grounds balance* by ensuring transnational engagement serves local priorities
- **Multi-Network Presence** (Partners) *enables coordination* by connecting slum communities globally for collective advocacy
- **Community Capacity Building** (Resources) *builds capacity* for residents to generate and use data about their own conditions
- **Climate Justice Integration** (Social Context) *grounds response* by connecting housing justice with climate adaptation

Condition interactions: Multi-Network Presence creates the transnational infrastructure. Democratization of Governance and Technology and Community Capacity Building enable resident-led data practices. Prioritizing Local Needs ensures global engagement serves local priorities. Climate Justice Integration connects housing and climate agendas.

Outcomes produced:

- **Resident-led urban development recognized by governments** (Impact): Democratization of Governance and Community Capacity Building combine for legitimate voice
- **Community-generated data empowering advocacy** (Impact): Community Capacity Building *builds capacity* for data generation
- **Community-led climate adaptation** (Impact): Climate Justice Integration *grounds response* in community priorities
- **Community-led development with transnational solidarity** (Balance): Prioritizing Local Needs *grounds balance* in local priorities with Multi-Network Presence *enabling* global connection
- **Transnational network amplifying community advocacy** (Balance): Multi-Network Presence *enables coordination* for amplified voice

Synthesis: Slum Dwellers International demonstrates how resident-led data practices, transnational networking, and climate justice integration combine to produce community-led urban development recognized by governments globally. The case shows that transforming urban governance requires community capacity for data generation and advocacy, transnational networks for solidarity and amplification, and framing that connects housing justice with climate adaptation—ensuring global engagement serves local priorities rather than imposing external agendas.

COMPACT EQUATION: For Slum Dwellers International, DOGAT (Governance) + PLNIGE (Org Info) + MP (Partners) + CCB (Resources) + CJI (Social Conte) = RUDR (Imp) + CDEL (Imp) + CCAI (Imp) + CDWT (Bal) + TNAC (Bal)

EQUATION 43/43: STOP LAPD SPYING COALITION

Location: USA | **Sector:** Police watchdog

Geographic Scope: Local (LA)

Maturity: Est. 2011; mature

Conditions and their mechanisms:

- **Resistance and Construction Integration** (Politics) *integrates resistance* by combining opposition to surveillance with building community capacity
- **Policy Influence and Systemic Change** (Politics) *directly produces* policy victories eliminating specific surveillance programs
- **Community Capacity Building** (Resources) *enables resistance* by building community skills to document, challenge, and defeat surveillance
- **Algorithmic Accountability and Transparency** (Technology Tools) *exposes systems* by mapping surveillance technology ecosystems
- **Inclusive Participation and Representation** (Social Context) *structures resistance* by ensuring directly-affected communities lead movement

Condition interactions: Inclusive Participation and Representation ensures affected community leadership—foundational for legitimacy. Community Capacity Building and Algorithmic Accountability and Transparency together enable documentation and exposure. Resistance and Construction Integration provides strategic orientation. Policy Influence translates into concrete victories.

Outcomes produced:

- **Shutdown of LAPD surveillance programs** (Democratic Practices): Policy Influence *directly produces* elimination of specific programs
- **Systemic mapping exposing surveillance ecosystems** (Democratic Practices): Algorithmic Accountability *exposes systems* comprehensively
- **Directly-affected community leadership** (Democratic Practices): Inclusive Participation *structures resistance* with affected leadership
- **Policy victories eliminating programs** (Impact): Policy Influence *directly produces* concrete change
- **Community capacity to challenge surveillance** (Impact): Community Capacity Building *enables* ongoing resistance

Synthesis: Stop LAPD Spying Coalition demonstrates how affected community leadership, capacity building, surveillance mapping, and strategic policy engagement combine to produce concrete victories against police surveillance systems. The case shows that effective accountability movements require affected community leadership for legitimacy, capacity building for sustained engagement, technical tools for documenting surveillance systems, and strategic policy engagement that translates exposure into elimination. The shutdown of LAPD's predictive policing programs represents a model for anti-surveillance movements nationally.

COMPACT EQUATION: For Stop LAPD Spying Coalition, RACI (Politics) + PIASC (Politics) + CCB (Resources) + AAAT (Technology) + IPAR (Social Conte) = SOLO (Dem) + SMES (Dem) + DCLI (Dem) + PVES (Imp) + CCTD (Imp)